They are

1s 2s 2p 3s 3p 3d

b.

S.No.	State	n	l	m_l	m_s	No. of states in shell
1	For 1s state	1	0	0	½	1
2	For 2s state	2	0	0	½	1
3	For 2p state	2	1	0, ± 1	½	3
4	For 3s state	3	0	0	½	1
5	For 3p state	3	1	0, ± 1	½	3
6	For 3d state	3	2	0, ± 1, ± 2	½	5

c. Now, number of states in shell having Quantum number $1s^2, 2s^2, 2p^6, 3s^2, 3p^6$, i.e., 18.

11. The density of aluminum is 2.70 g/cm³ and its molecular weight is 26.98 g/mole. (a) Calculate the Fermi energy. (b) If the experimental value of E_F is 12 eV, what is the electron effective mass in aluminum? Aluminum is trivalent. **[TU Microsyllabus, P 24.8]**

Solution:

Density of aluminum (ρ) = 2.70 gm cm^{-3}

Molecular weight of Aluminum (m) = 26.98 gm mole^{-1}

a. Fermi energy (E_F) = ?

b. Fermi energy (E_F) = 12 eV = $12 \times 1.6 \times 10^{-19}$ J

Effective mass (m_e^*) = ?

We know hat

Number of free electrons per unit volume

$$N = \frac{v\rho N_A}{m}$$

i.e. $$N = \frac{3 \times 2.70 \times (6.02 \times 10^{23})}{26.98} = 1.807 \times 10^{23} \text{ electrons cm}^{-3}$$

$$= 1.807 \times 10^{29} \text{ electrons m}^{-3}$$

Now, we have relation

$$E_F = \frac{\hbar^2}{2m_e}(3N\pi^2)^{2/3} = \frac{\left(\frac{6.626 \times 10^{-34}}{2\pi}\right)^2}{2 \times 9.1 \times 10^{-31}}(3 \times 1.807 \times 10^{29} \times \pi^2)^{2/3}$$

$$= 1.867 \times 10^{-18} \text{ J} = \frac{1.867 \times 10^{-18}}{1.6 \times 10^{-19}} \text{ eV}$$

$\therefore$ $E_F = 11.66$ eV

b. Again, for m_e^*

$$E_F' = \frac{\hbar^2}{2m_e}(3N\pi^2)^{2/3}$$

$$m_e^* = \frac{\hbar^2}{2E_F'}(3N\pi^2)^{2/3}$$

TRIBHUVAN UNIVERSITY

Institution of Science and Technology

Course Title: Physics (PHY113) | Full Marks: 60
Semester: I | Pass Marks: 24
Duration: 3 Hours | Credit Hour: 3

TU QUESTIONS-ANSWERS 2077

Attempt any TWO questions. **[10×2 = 20]**

1. **Explain RTL and TTL gates. How memory and clock circuits can be made by using these gates? Explain how they work?**

Solution:

Resistor transistor logic (RTL) gate family is often found in an IC. In which all the logic are implemented using resistor and transistors.

An example of

RTL gate is shown in figure, which implements the NOR function.

The circuit consists of two inverters wired in parallel with V_{out} and the collector R_C common to both. V_A and V_B are two inputs and each affects the output of their respective transistor.

Suppose $V_A = 5V$ and $V_B = 0V$, transistor A will conduct, that is, I_{B_1} will be high and therefore I_C will be high. Consequently, the collector of A will be at approximately 0V with respect to ground ($V_{CE} \approx 0V$) and V_{out} will be 0V.

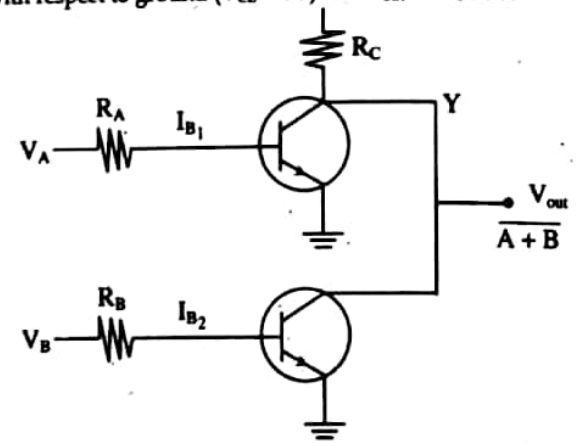


Figure: RTL circuit for NOR logic function.

A similar output will be obtained if $V_A = 0V$ and $V_B = 5V$ or if both $V_A = V_B = 5V$.

Thus, if either or both inputs are logic level 1, the output will be at logic level 0.

If $V_A = V_B = 0V$, then neither of the two transistor will conduct ($I_C = 0$) and both collectors will be at 5V with respect to their emitters, which are grounded; $V_{out} = 5V$.

Thus the circuit implements the NOR logic function.

The primary advantage of RTL technology is that it uses a minimum number of transistors.

TTL gates have many advantages over RTL

An example of

TTL gate is shown in figure, which implements a NAND function.

The input transistor Q_1 is an npn transistor with several emitters. Transistor with 8 or more input emitters are not unusual nor difficult to manufacture in IC form.

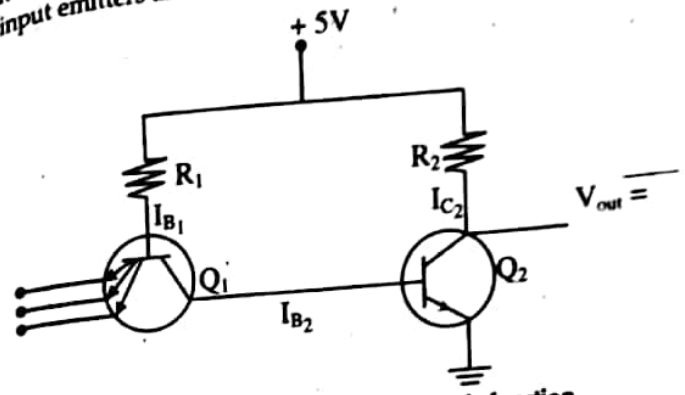


Figure: TTL gate for NAND logic function.

If anyone the inputs of Q_1 is grounded (logic level 0), the base emitter junction of Q_1 will be forward based, so a large base current I_{B1} that turns Q_1 on. As a result, no current flows in the emitter base junction of Q_2. With Q_2 cutoff, its collector emitter voltage becomes 5V, that is, $V_{out} = 5V$.

Thus, if any of the inputs are at logic level 0, the output will be at logic level 1.

If all the input of Q_1 are at 5V, its base emitter will be reversed biased so no current can flow from the base of Q_1 to its emitter, current can flow through R and through the forward biased base collector of Q_1. This will turn Q_2 on and make its collector emitter voltage 0V therefore $V_{out} = 0V$.

Thus, when all inputs are at logic level 1, the output will be a logic level 0.

This is logically a NAND gate.

The electronic circuit whose output remains as set until something is done to change is called memory circuit. Logic circuits that incorporate memory cells are called sequential logic circuits.

The basic memory circuit is called the flip-flop.

A flip-flop is a bistable memory element widely used in sequential logic circuits. It has two stable outputs Q and its complementary value $\bar{Q}$.

When Q = 0 and $\bar{Q} = 1$, the flip-flop is in reset state

When Q = 1 and $\bar{Q} = 0$, the flip-flop is in set state.

The flip-flop is often called a latch, since it will hold, or latch, in either stable state.

RS flip-flop can be constructed by cross coupling of two NOR gates as shown in figure 1.

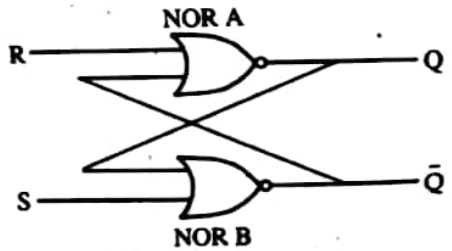


Figure 1: RS flip-flop using NOR gates

There are two inputs to the flip-flop defined as R and S and has two outputs Q and $\bar{Q}$. The input/output possibilities for this RS flip-flop are discussed below:

1. **When R = 0 and S = 0:** Since the NOR gate is high sensitive. So a '0' at the input of a NOR gate has no effect on its output, the flip-flop simply remains in its present state: that is, Q remains unchanged.
2. **When R = 0 and S = 1:** In this input condition, the output of NOR gate B is low. Both inputs to NOR gate A are now low, so the output must be high. Thus a 1 at the S input is said to SET the flip-flop, and it switches to the stable state where Q = 1 (or $\bar{Q}$ = 0)
3. **When R = 1 and S = 0:** This condition forces the output of NOR gate, a low, and since both inputs to NOR gate B are low, the output must be high. Thus a '1' at the R input is said to RESET the flip-flop, and it switches to the stable state where Q = 0 (or $\bar{Q}$ = 1).
4. **When R = 1 and S = 1:** This is a forbidden condition, as it forces the outputs of both NOR gates to the low state. Which violates the basic definition of a flip-flop that requires Q and $\bar{Q}$ be complement to each other. If this condition is for some reason imposed, the resulting state of Q is not predictable.

Truth table for NOR-gate RS flip-flop.

R	S	Q	Action
0	0	Last state	No change
0	1	1	SET
1	0	0	RESET
1	1	?	Forbidden

A clock is a circuit that produces periodic pulses. Most sequential logic circuits are driven by a clock signal. A clock signal is a particular type of signal that oscillates between a high and a low state. A clock can be constructed with a multivibrator. Multivibrator is an electronic circuit that generates square waves or other non-sinusoidal waves such as rectangular or saw tooth wave.

The astable or free running multivibrator generates square waves output without any external exutation. It has no stable state. It has only two quasistable states between which it keeps on oscillating.

Figure (1) shows the circuit of an astable multivibrator consisting two identical transistor Q_1 and Q_2. It is a two stage RC coupled amplifier. The output of each stage is coupled to the input of the other through a capacitor (C_1 or C_2).

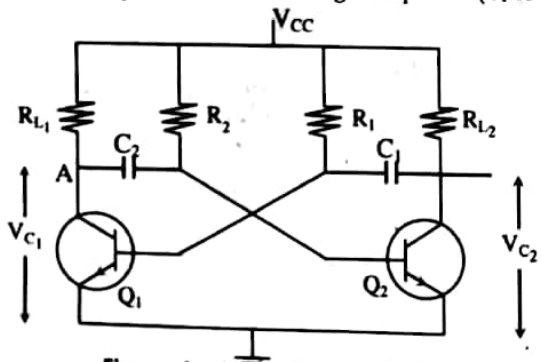


Figure : Astable transistor multivibrator.

When power V_{CC} is applied, then collector current starts flowing in Q_1 and Q_2. The coupling capacitors C_1 and C_2 start charging up.

The characteristics of two similar transistor can be exactly a like.

Suppose that Q_1 starts conducting before Q_2. The feedback system in such that Q_1 will be very rapidly driven to saturation and Q_2 to cut off. The circuit operation may be explained as follows.

1. Since Q_1 is in saturation whole of V_{CC} drops across R_{L_1}. Hence V_{C_1} = 0 and point A is at zero or ground potential.
2. Since Q_2 is in cut off i.e. it conducts no current, there is no drop across R_{L_2}. Hence point B is at V_{CC}.
3. Since A is at 0V, C_2 starts to charge through R_2 towards V_{CC}.
4. Capacitor C_2 now begins to discharge which decreases the reverse bias on base of Q_2. Therefore Q_2 begins to conduct. Consequently, collector of Q_2 becomes less positive. This negative going voltage signal is applied to base of transistor Q_1 through the capacitor C_1. Which results, Q_1 to cutoff simultaneously Q_2 to saturation. Now V_{C_2} decreases to zero when Q_2 gets saturated. Consequently, potential at B decrease from V_{CC} to almost zero.

The transistor Q_1 remains cut off and Q_2 in conduction until capacitor C_1 discharges through R_1, enough to decrease the reverse bias of Q_1.

The whole cycle repeats.

2. **Set up differential equation for an oscillation of a spring using Hooke's and Newton's second law. Find the general Solution: of this equation and hence the expressions for period, velocity and acceleration of oscillation.[10]**

Ans: [See Model Question No. 2]

3. **Describe Frank-Hertz experiment. Interpret how the results of this experiment advocate atomic model proposed by Bohr?**

Soluton:

By postulating the existence of a discrete spectrum of energy levels through the angular momentum condition i.e. L = mvr = nh. Bohr was able to predict the correct electromagnetic spectrum for hydrogen. The existence of discrete energy levels in atoms was demonstrated directly by James Franck and Crustav Hertz in 1914.

A schematic of the experimental set up is shown in figure (1)

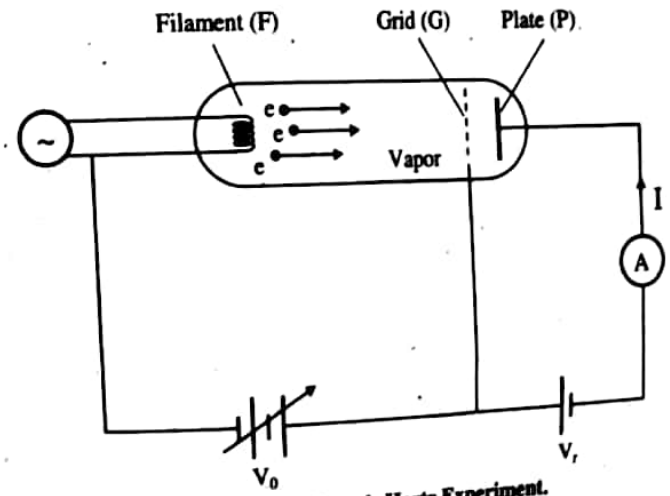


Figure 1: Franck-Hertz Experiment.

The essential part of the apparatus consists of a tube containing vapour of the element under study (Franck and Hertz had studied mercury vapour). The tube contains three electrodes: a filament (F) that provides electrons when heated, a plate (P), and a grid (G). A grid is a charged screen that can attract or repel electrons. But due to the open space, the majority of the electron pass through it. A variable accelerating potential V_0 is applied between the filament and the grid and small retarding voltage V_r (~1V) is applied between the plate and the gird. The electrons will reach the grid with a kinetic energy $E_k = eV_0$. After reaching the grid majority, these electrons will go through the open space and contribute to the plate current I.

If $V_r > V_0$, the electrons will be turned back before they can reach the plate and they will not contribute to the current measured by Ammeter.

But even if $V_r < V_0$, the electron will not able to reach the plate if they lose enough kinetic energy through collision with atoms in the tube as they travel between the filament and the grid.

In the absence of any vapour the I–V characteristics is shown in figure (2) by dashed line

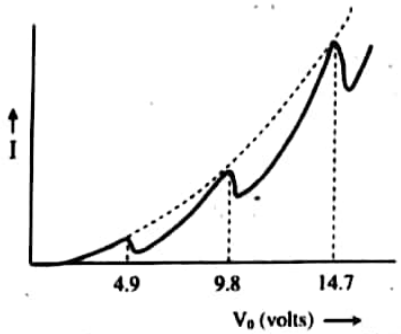


Figure 2: Dependence of plate current with V_0

The solid curve represents the effect of mercury present in the tube on increasing the accelerating potential. As a result, plate current also increases. As V_0 reaches the value of a critical potential 4.9 V, an electron acquires 4.9 eV of energy on reaching G. The electron loses all its energy in an inelastic collision with a mercury atom. Thus, the electron is left with no energy to reach P. Consequently the current drops abruptly. This suggest that the atom has absorbed this 4.9 eV energy to raise it from the ground state to a state of higher energy. This current dipping to a minimum at 4.9 V does not reach zero because statistically some electron may succeed in reaching the collector plate, avoiding an inelastic collision with a mercury atom. Then as V_0 is further increased, the electrons start reaching P after the inelastic collision, if they are left with energy to overcome V_r. Hence, the plate current again increases. At V = 9.8V, the current again dips to a second minimum. This can be explained if an electron of 9.8 eV energy suffers two consecutive inelastic collision with different mercury atom before it reaches the plate. Such an electron excites both the mercury atoms to their first excited states, losing 4.9eV energy in each collision. This explains the second minimum.

ch time there is an inelastic collision, the mercury atoms will be excited and turn to the ground state by the emission of photons. There should be a spectral e whose frequency is given by hf = 4.9 eV or λ = 2530 Å. Such a wave length is und in the spectrum of Hg.

n energy diagram for Hg from spectral data in shown in figure (3)

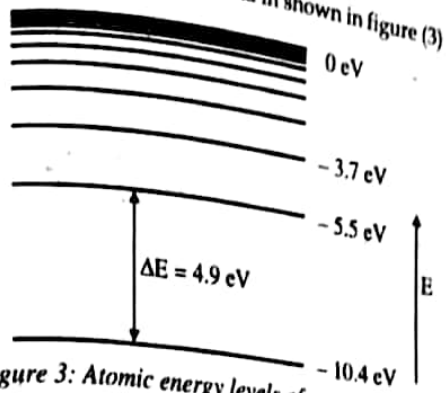


Figure 3: Atomic energy levels of mercury atom

The energy difference ΔE between the first excited state and ground state is

$$\Delta E = 10.4\text{ eV} - 5.5\text{ eV} = 4.9\text{ eV}.$$

The energy difference ΔE between the second excited state and ground state is ΔE = 10.4 eV – 3.7eV = 6.7 eV. But we may not observe a dip when V_0 = 6.7V. This is because the number of electron that are able to avoid the 4.9 eV collision and gather 6.7 eV is a small fraction of the total number of electron. Which reduces the magnitude of 6.7 eV dip and makes it more difficult to observe. This can be observed if the sensitivity of the experiment is increased.

Thus, this experiment shows that the energy lost by the electron in its inelastic collision with the mercury atom reappears as a quantum of energy of wave length hc/E. This experiment shows in a very convincing way the existence of discrete energy levels in the mercury atom. While it isn't evident in the original measurements of the figure, this series of dips in current at approximately 4.9 volt increments continues to potentials of at least 70 volts.

Group B

Short Answer Questions [8×5=40]

4. Discuss magnetic dipole moment. What is its effect on atom and on molecules? Explain.

Soluton:

The magnetic dipole moment is defined as the product of current through the loop and the area of loop.

$$\mu = IA$$

The expression for torque can now be written as

$$\tau = \mu B \sin\theta \text{.....(3)}$$

Equations (2) and (3) give the magnitude of the torque but they do not specify the direction of τ. The direction of the torque can be obtained by expressing equation (3) in vector form

$$\tau = \vec{m} \times \vec{B} \text{ .....(4)}$$

TRIBHUVAN UNIVERSITY

Institution of Science and Technology

Course Title: Physics (PHY113)
Semester: I
Duration: 3 hours

Full Marks: 60
Pass Marks: 24
Credit Hour: 3

TU QUESTIONS-ANSWERS 2078

Attempt any TWO questions. **[10×2 = 20]**

Discuss electrical conductivity of semiconductors. Derive expression for conductivity in terms of impurity ionization energy. Give a plot to discuss "Theoretical temperature dependence of the electrical conductivity of an impurity semiconductor".

Solution:

When an electric field is applied to the semiconductor valance holes drift in the direction of applied field where as conduction electron drift in the direction opposite to the applied field. The drifting of both electrons in conduction band and holes in valence band contribute to current in one direction. Therefore the total current density in semiconductor can be written as:

$$J = n e v_{de} + p e v_{dh}$$

Where, n = electrons concentration in conduction band
p = holes concentration in valence band
v_{de} = drift velocity of electron
v_{dh} = drift velocity of holes.

Since, $\mu = \frac{v_d}{E} \Rightarrow v_d = \mu E$

$$\therefore J = n e \mu_e E + pe \mu_h E$$
$$\sigma E = (n e \mu_e + pe \mu_h) E$$
$$\sigma = n e\mu_e + p e \mu_h$$

Where μ_e and μ_h are electrons and hole mobilities

$\mu_e = \frac{e \tau_e}{m_e^*}$, $\mu_h = \frac{e \tau_h}{m_h^*}$; τ_e and τ_h represents the mean time between the collisions with ions for electrons and holes respectively

m_e^* and m_h^* are the effective masses of electrons and holes respectively

n - type semiconductor

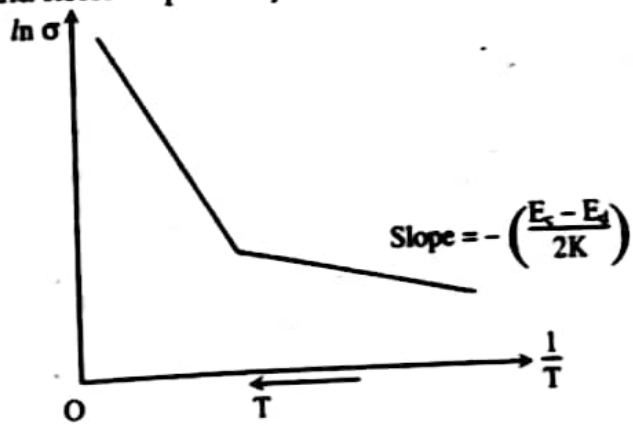
The conductivity of n - type semiconductor is mainly due to the electrons in the conduction band and is given by

$$\sigma = ne \mu_e$$

Now,

$$n_e = \frac{N_d}{1 + e^{\frac{E_f - E_d}{KT}}}$$

Therefore,



Dependence of electric conducitivity with temperature for an impurity semiconductor

$$\sigma = e\mu_e N_d \frac{1}{e^{\frac{E_f - E_d}{KT}}}$$

or $$\ln \sigma = -\ln\left[e^{\frac{E_f - E_d}{KT}} + 1\right] + C_1$$

where C_1 is a constant given by $C_1 = \ln e\mu_e N_d$, N_d is the number of donor atoms per unit volume.

When a graph is plotted between $\ln\sigma$ and $\frac{1}{T}$ a straight line for high T and a flatter line for smaller T as shown in figure (2).

At low temperature ($E_f > E_d$), $\ln \sigma$ vary linearly with $\frac{1}{T}$ and the slope will be $-\frac{(E_f - E_d)}{2K}$

for, $E_f = \frac{E_c + E_d}{2}$ at low T i.e. slope $= -\left(\frac{E_c - E_d}{2K}\right)$

At high temperature (as $E_d - E_f >> KT$) intrinsic conductivity begins to dominate the expression.

p – type semiconductor

The conductivity of a p – type semiconductor is primarily due to the holes in the valence band and is given by $\sigma = n\,p\,\mu_h$

Now, $$p = \frac{N_a}{e^{\frac{E_a - E_f}{KT}} + 1}$$

$$\therefore \quad \sigma = e\,\mu_h N_a \frac{1}{e^{\frac{E_a - E_f}{KT}} + 1}$$

or $$\ln \sigma = -\ln\left[e^{\frac{E_a - E_f}{KT}} + 1\right] + C_2$$

Where C_2 is a constant $= \ln e\mu_h N_a$, N_a is the number of acceptor atoms per unit volume at acceptor level.

If we plot $\ln \sigma$ against $\frac{1}{T}$, $\ln \sigma$ behaves similar to that of an n – type semiconductor.

2. Describe processes involved in the fabrication of integrated circuits include epitaxial growth, oxidation, oxide removal and pattern definition, doping (impurities in the Si) and interconnection of components.

Solution:

The first process involved in the fabrication of integrated circuit is photolithography which includes epitaxial growth, oxidation, oxide removal and pattern definition. The photolithography is followed by doping (introduction of selective impurities in Si) and metallization (Interconnection of components).

Photolithography

It is the process which involves photographic transfer of a pattern to the surface of wafer to make diffusion window by etching. In this process a geometrical pattern

11. The energy gap in Silicon is 1.1 eV, whereas in diamond it is 6 eV. What conclusion can you draw about the transparency of the two materials to visible light (4000Å to 7000Å). **[TU Microsyllabus, P 25.16]**

Solution:

Let λ_C be the wavelength at which the absorption by the material changes abruptly, we have

$$E_g = \frac{hC}{l_c}$$

1. For Silicon, $Eg = 1.1 \times 1.6 \times 10^{-19}$ J

$$1.1 \times 1.6 \times 10^{-19} = \frac{6.62 \times 10^{-34} \times 3 \times 10^{8}}{l_C}$$

$\therefore$ $\lambda_c = 1.128 \times 10^{-6}$ m > $(4 \times 10^{-7}$ m)

2. For diamond, $Eg = 6 \times 1.6 \times 10^{-19}$ J

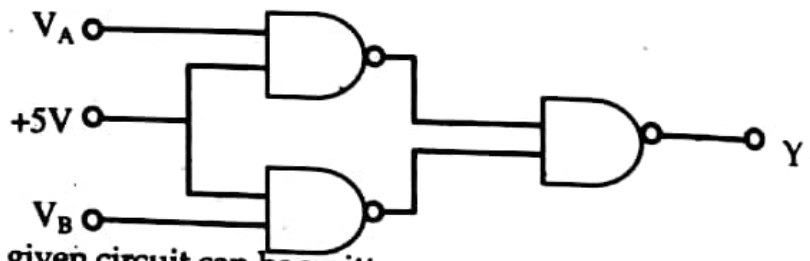
$$6 \times 1.6 \times 10^{-19} = \frac{6.62 \times 10^{-34} \times 3 \times 10^{8}}{l_C}$$

or $\lambda_C = 2.068 \times 10^{-7}$ m < $(4 \times 10^{-7}$ m)

Hence the critical λ for Silicon & diamond are 1.28×10^{-6} m & 2.06×10^{-7} m respectively. Thus, the Silicons is opaque to visible light where as the diamond is transparent to visible light.

12. Find the truth table for the circuit shown in figure. What logic function does the circuit perform? What logic function will the circuit perform if the constant + 5V input to the first two gates are changed to ground potential? **[TU Microsyllabus, P 27.9]**

Solution:



a. The output of the given circuit can be written as:

$$Y = [\overline{(5V.V_A)} \,.\, \overline{(5V.V_B)}] = V_A + V_B = \text{OR – operation}$$

Truth table

Inputs			Outputs				
V_A	V_B	+5V	$5V.V_A$	$5V.V_B$	$\overline{5V.V_A}$	$\overline{5V.V_B}$	$Y=\overline{(5V.V_A)} \,.\, \overline{(5V.V_B)}$
0	0	+5	0	0	+5	0	0
0	5	+5	0	5	+5	5	5
5	0	+5	5	0	0	0	5
5	5	+5	5	5	0	5	5

b. If the constant + 5V input to the first two gates is changed to ground potential i.e., (+0V) then,

$$\text{Output } Y = \overline{(0V.V_A)} \,.\, \overline{(0V.V_B)}$$

$$= \overline{5V.5V}$$

$$Y = 0V$$

Hence, none of the logic function perf...

Because a magnetic dipole experiences a torque when placed in an external field, work must be done to change it's orientation. This work done is also be referred as energy of dipole

$$U = \int_{\theta_i}^{\theta_f} \tau d\theta = \int_{\theta_i}^{\theta_f} \mu B \sin\theta \, d\theta = \int_{90^\circ}^{\theta} \mu B \sin\theta \, d\theta$$

$$= -\mu B \cos\theta \text{.....(5)}$$

This can be expressed in dot product as

$$U = -\vec{m} \cdot \vec{B} \text{ .....(6)}$$

From equation (5) we conclude that,

$U_{max.} = \mu B$, when $\theta = \pi$, that is when μ and B are anti-aligned

$U_{min.} = -\mu B$, when $\theta = 0$, that is when μ and B are aligned.

Electrons revolving around atomic nuclei, electrons spinning on their axes, and rotating positively charged atomic nuclei all are magnetic dipoles. The sum of these effects may not be a magnetic dipole. If they do not fully cancel , the atom is a permanent magnetic dipoles, as iron atoms. The same alignment to form ferromagnetic domain also constitute a magnetic dipole

5. Describe the following process of IC production: (a) Oxidation, (b) Pattern definition, and (c) Doping

Solution:

[See T.U. Model No. 6]

6. Explain the construction and bipolar junction transistor(BJT).

Solution:

[See T.U. 2075 Question No. 6]

7. A roulette wheel with I = 0.5 kg-m²rotating initially at 2 rev/sec coasts to stop from the constant friction torque of the bearing. If the torque is 0.4 N-m, how long does it take to stop?

Solution:

Moment of inertia of roulette wheel (I) = 0.5 kg m²

Initial speed of revolution per second (f_0) = 2 rev s^{-1}

Time taken to stop (t) = ?

Now,

$$\tau = I\alpha = I\frac{(\omega_f - \omega_0)}{t}$$

$$0.4 = \frac{0.5 \times (0 - 2\pi \times 2)}{t} = \frac{0.5 \times 4\pi}{t}$$

$$t = 15.77 \text{ sec}$$

8. Two large parallel plates are separated by a distance of 5 cm. The plates have equal but opposite charges that create an electric field in the region between the plates. An α particle ($q = 3.2 \times 10^{-19}$C, $m = 6.68 \times 10^{-27}$ kg) is released from the positively charged plate, and it strikes the negatively charged plate 2×10^{-6} sec later. Assuming that the electric field between the plates is uniform and perpendicular to the plates, what is the strength of the electric field?

Solution: [See Model Questions No. 8] **[5]**

9. A neutron spectroscopy a beam of mono energetic neutrons is obtained by reflecting reactor neutrons from a beryllium crystal. If separation between the atomic planes of the beryllium crystal is 0.732 Å, what is the angle between the incident neutron beam and the atomic planes that will yield a monochromatic beam of neutrons of wavelength 0.1 Å. **[TU Microsyllabus, P 19.11]**

Solution:

Separation between the atomic plane (d) = 0.732Å = 0.732×10^{-10} m

Wavelength of monochromatic beam of neutrons (λ) = 0.1Å = 0.1×10^{-10} m

Angle between the incident neutron beam and the atomic planes (θ) = ?

From Bragg's law,

$$2d \sin\theta = n\lambda$$

or $\sin\theta = \frac{n\lambda}{2d}$ [Since, n = 1, for first order]

or $\theta = \sin^{-1}\left(\frac{\lambda}{2d}\right)$

or $\theta = \sin^{-1}\left(\frac{0.1 \times 10^{-10}}{2 \times 0.732 \times 10^{-10}}\right)$

$\therefore \theta = 3.92^\circ$

10. What is the probability of finding a particle in a well of width 'a' at a position $\frac{a}{4}$ from the wall if n = 1, if n = 2, if n = 3. Use the normalized wave function, $\psi(x, t) = \sqrt{\frac{2}{a}} \sin\frac{n\pi x}{a} e^{-iEt/\hbar}$. **[TU Microsyllabus, P 20.12]**

Solution

Width of well = a

Position (x) = $\frac{a}{4}$

Normalized wave function $\psi(x, t) = \sqrt{\frac{2}{a}} \sin\frac{n\pi x}{a} e^{-iEt/\hbar}$

We know

Probability of finding a particle, $P = \psi^*\psi$

At $x = \frac{a}{4}$

$$P = \left(\sqrt{\frac{2}{a}} \sin\frac{n\pi}{a} \cdot \frac{a}{4} e^{+iEt/\hbar}\right)\left(\sqrt{\frac{2}{a}} \sin\frac{n\pi}{a} \cdot \frac{a}{4} e^{-iEt/\hbar}\right)$$

$$P = \frac{2}{a} \sin^2\frac{n\pi}{4} \text{ .....(1)}$$

If n = 1, $P_1 = \frac{2}{a}\sin^2\frac{\pi}{4} = \frac{2}{a}\left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{a}$ For n = 2, $P_2 = \frac{2}{a}\sin^2\frac{2\pi}{4} = \frac{2}{a}$

And for n = 3, $P_3 = \frac{2}{a}\sin^2\frac{3\pi}{4} = \frac{2}{a}\sin^2 135^\circ = \frac{2}{a}\left(\frac{1}{\sqrt{2}}\right)^2$

$\therefore P_3 = \frac{1}{a}$

Hence, the probability of finding a particle in a well of width of a position $x = \frac{a}{4}$ form the wall for n = 1, n = 2, and n = 3 are $\frac{1}{a}, \frac{2}{a}$ and $\frac{1}{a}$ respectively.

transferred from a mask to the surface of silicon wafer. Lithography literally means "Writing on stone". Photolithography involves following steps.

Step i: Coat Si with oxide then with photoresist (Oxidation)

At first the Silicon single crystal is oxidized in an oxidation furnace to form a thin layer of SiO_2, which is excellent barrier against diffusion. The oxide layer is grown by heating the silicon wafer to temperatures ranging between 1000°C and 1200°C in an atmosphere of either pure oxygen or steam. The thickness of the oxide layer depends on the oxidation time and the temperature and the composition of the atmosphere in which the oxidation is performed. By careful selection of these three parameters, the exact thickness of the layer can be controlled. A layer 0.1 m thick can be grown in one hour at T = 1000°C, in pure oxygen. In the same time, a layer 0.5 μm thick grows in a steam environment.

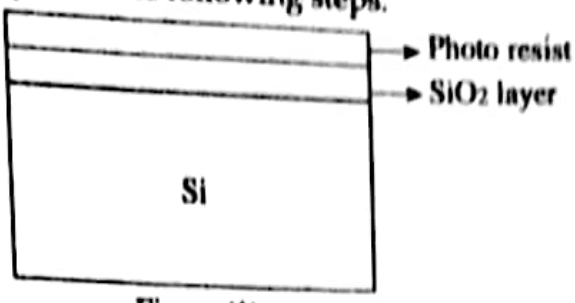


Figure (1)

Again coat the wafer with a radiation sensitive polymer film called the photo resist. Spin the silicon wafer very fast so that coating is uniform (figure 1).

Step ii: *Expose to radiation and develop the pattern (Pattern definition)*

Allow the UV radiation to fall on photo resist through mask. A mask is a glass plate with transparent and opaque regions made on it. Only those regions of the mask which are transparent allow the radiation to fall on the semiconductor. Only those portions which are exposed to radiation, their properties are going to change. The photo resists from the exposed regions are removed. Now the mask pattern is transferred to top of wafer. (Figure 2)

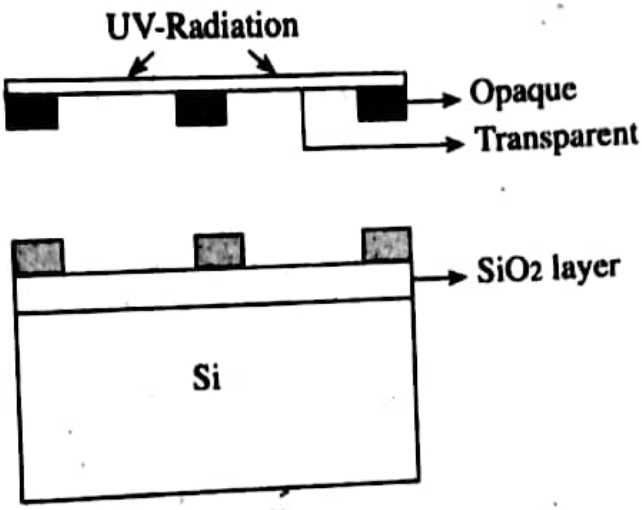


Figure (2)

Step iii: *Oxide removal*

Now the substance is kept in diluted etching solution (hydro fluoric acid, HF), the SiO_2 from the Si regions corresponding to transparency of mask is removed. (Figure 3)

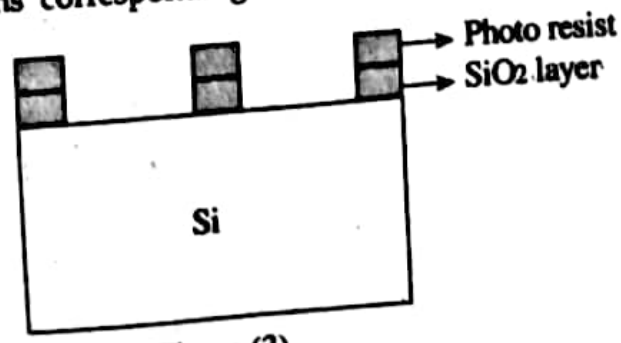


Figure (3)

Step iv: All the photoresist is removed by keeping it on photoresist remover solution (Figure 4).

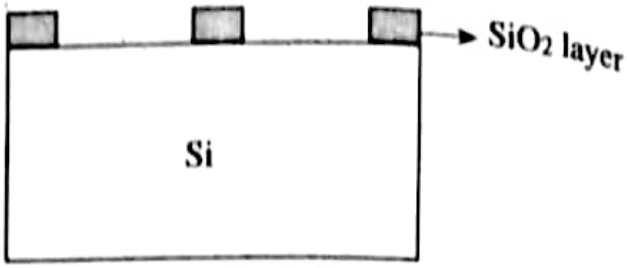


Figure (4)

Compare the pattern on mask with pattern of SiO_2 on the top of Si wafer, the opaque regions on the mask have corresponding pattern of SiO_2 on Si wafer. The regions corresponding to transparency of mask have no oxide. These are the regions where the dopant will be incorporated. This process in which UV light is used to produce the diffusion window pattern is called *Photolithography*.

The pattern on the mask and wafer are identical. Those pattern obtained on wafer are said to be due to positive photoresist. A positive resist is one which gets soften when exposed to radiation. If the photoresist is hard, the complement pattern is obtained with semiconductor regions etched with UV radiation. It is called a negative photo resist.

The fabrication of IC is done plane by plane at one surface of the wafer so that the deepest region tends to be fabricated first. We will consider fabrication of anpn bipolar junction transistor. The fabrication follows following steps.

1. We start with a p-type single crystal of thickness about 200 μm, oriented {1, 1, 1} and with a resistivity of 10Ω-cm.

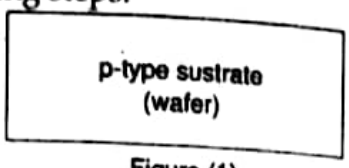


Figure (1)

2. Grow an n-type epilayer on the top of wafer by epitaxy [epitaxy - arranged upon]. The epitaxial layer is very much thinner to the original layer of bulk. This grown epitaxial layer is going to be collector of npn transistor.

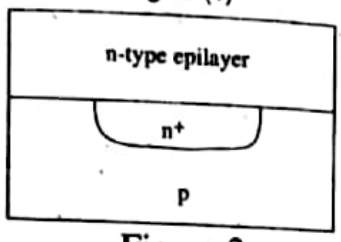


Figure 2

3. Now doping acceptors over n-region to form p-region for base diffusion [after photolithography] (figure 3)

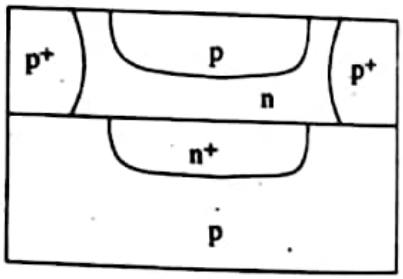


Figure 3

4. Again dop donors to form emitter. (After photolithography)

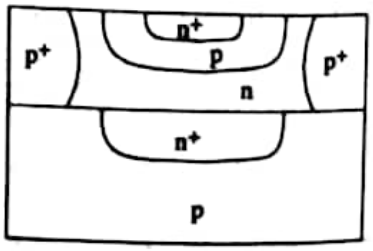


Figure 4

Metallization involves deposition of metal (mostly aluminum) elements over base, collector and emitter region (After photolithography) for electrical connection. Remove all oxide layer (used in photolithography) and unwanted metal deposition by etching. Thus fabricated 'npn' transistor can be used after electrically tested.

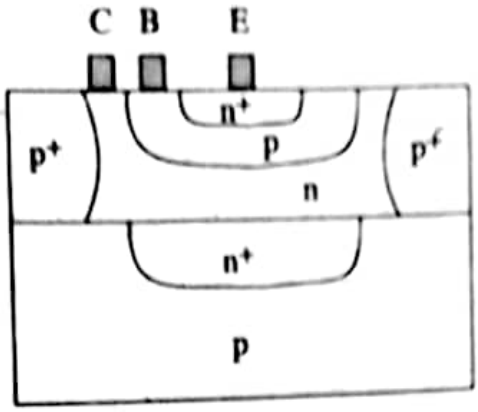


Figure 5

The steps needed to build a diode are identical to those used in the fabrication of a transistor except that the last diffusion of donor impurities (step 4) is omitted.

Find expression for force on a current carrying wire in a magnetic field to find the force experienced by a single charge.

Solution:

Force on a Moving Charge

Magnetic force is due to the motion of electric charges (i.e. electric current). A magnetic force exists between two charges if and only if both charges are moving relative to an observer.

Experimentally it is found that,

The magnitude of the magnetic force exerted is directly proportional to the amount of charge q.

The magnetic force depends upon the velocity of the charged particle (v).

The magnetic force depends upon the magnetic field (B)

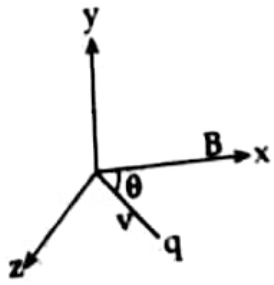
The magnetic force is proportional to sine of angle between particle velocity and magnetic field.

Therefore, Magnetic force is given by

$$F = q\,v\,B\sin\theta$$

or, $$\vec{F} = q(\vec{v} \times \vec{B})$$

This shows that, magnetic force always acts perpendicular to the magnetic field.

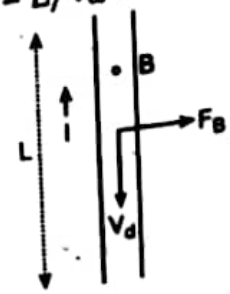


Magnetic force act only when the particle is in motion.

The magnitude of magnetic field is defined by the relations.

$$B = \frac{F}{qv\sin\theta}$$

Consider a conducting wire of length L. A current 'I' is flowing in upward direction. The conduction electrons cross this length of conductor in time $t = L/v_d$, where v_d is drift velocity of electron.



The amount of charge moved through conductor in this time, $q = I\,t \Rightarrow q = \frac{IL}{v_d}$

The magnetic force for this amount of charge, $F_B = B\,q\,v_d \sin\theta$ where θ is angle between $\vec{B}$ and $\vec{v}_d$. In our case the magnetic field is directed perpendicularly to page and v_d along the conductor. That is $\theta = 90°$

Therefore, $F_B = Bq\,v_d = B.\frac{IL}{vd}.\,v_d = ILB$

$$F_B = ILB$$

If the magnetic field is not perpendicular to wire, the magnetic force on a conductor is given by

$$F_B = ILB \sin\phi$$

In vector form, $\vec{F}_B = I(\vec{L} \times \vec{B})$ .....(1), where ϕ is the angle between $\vec{L}$ and $\vec{B}$

This equation is equivalent to, $\vec{F}_B = q(\vec{v} \times \vec{B})$ which is magnetic force on a moving charge.

Group B

Attempt any EIGHT questions. **[5×8=40]**

4. Give spectrum of Hydrogen atom and discuss its lines.

Solution:

The energy levels of the Bohr atom are given by,

$$E_{n_2} = \left(\frac{-me^4}{8\varepsilon_0^2 h^2}\right)\frac{1}{n_2^2} \qquad E_{n_1} = \left(\frac{-me^4}{8\varepsilon_0^2 h^2}\right)\frac{1}{n_1^2}$$

Where E_{n_2} and E_{n_1} are the total energy of electron in the outer orbit of quantum number n_2 and total energy of electron in the inner orbit of quantum number n_1 respectively.

When the electron makes a transition from n_2^{th} orbit to the n_1^{th} orbit, the frequency f of the radiation emitted is given by

$$hf = E_{n_2} - E_{n_1} = \left(\frac{-me^4}{8\varepsilon_0^2 h^2}\right)\frac{1}{n_2^2} + \left(\frac{me^4}{8\varepsilon_0^2 h^2}\right)\frac{1}{n_1^2} = \left(\frac{me^4}{8\varepsilon_0^2 h^2}\right)\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$$\therefore \quad f = \frac{me^4}{8\varepsilon_0^2 h^3}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

Let λ be the wave length of radiation. Then $f = \frac{c}{\lambda}$, where c is the speed of light in vacuum.

$$\text{or,} \quad \frac{c}{\lambda} = \frac{me^4}{8\varepsilon_0^2 h^3}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$$\text{or,} \quad \frac{1}{\lambda} = \frac{me^4}{8\varepsilon_0^2 c\,h^3}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$$\therefore \quad \frac{1}{\lambda} = R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

Where $R = \frac{me^4}{8\varepsilon_0^2 c\,h^3}$ is the Rydberg constant $= 1.097 \times 10^7\ m^{-1}$.

is value is nearly the same as the experimental value for the hydrogen atom. above equation is called Bohr's equation. The emission of all the spectral series lines in the hydrogen spectrum are explained as follows:

Lyman series (in ultra-violet region, invisible)

This series of lines is emitted by transition of the electron from outer orbits of quantum number, $n_2 = 2, 3, 4....$ to the first orbit of quantum number $n_1 = 1$.

The wavelength of this series of lines are given by, $\frac{1}{\lambda} = R\left(\frac{1}{1^2} - \frac{1}{n_2^2}\right)$, where $n_2 = 2, 3, 4, ...$

Balmer Series (Visible): This series of lines is emitted by transitions of the electron from outer orbits of quantum number $n_2 = 3, 4, 5$ to the second orbit of quantum number $n_1 = 2$. Thus, the wavelength of this series of lines are given by

$$\frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{n_2^2}\right); \text{ where } n_2 = 3, 4, 5, ...$$

The lines are also called H_α, H_β, H_γ and H_δ lines for $n_2 = 3$, $n_2 = 4$, $n_2 = 5$ and $n_2 = 6$ respectively.

Paschen series (in infrared region, invisible)

For this series $n_1 = 3$, and $n_2 = 4, 5, 6 .....$

Brackett series (in infrared region, invisible)

For this series $n_1 = 4$, and $n_2 = 5, 6, 7,....$

Pfund series (in far infrared region, invisible)

For this series $n_1 = 5$, and $n_2 = 6, 7, 8,....$

The transitions responsible for some of the wavelength in each of the spectral series of hydrogen spectrum are shown in figure.

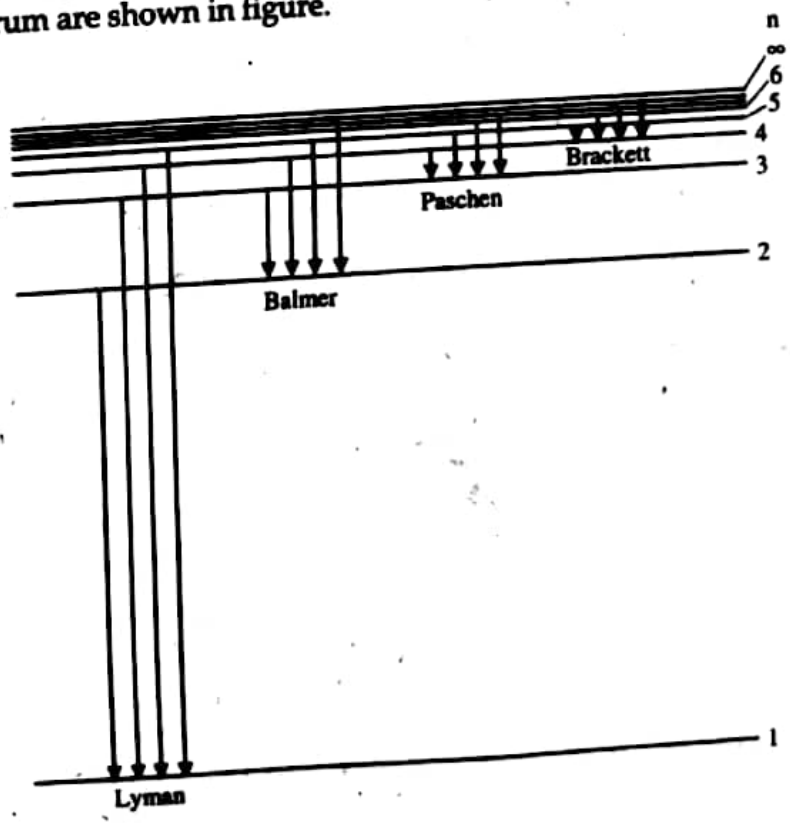


Figure: Energy levels and spectra of hydrogen

They are

1s 2s 2p 3s 3p 3d

b.

S.No.	State	n	l	m_l	m_s	No. of states in shell
1	For 1s state	1	0	0	½	1
2	For 2s state	2	0	0	½	1
3	For 2p state	2	1	0, ± 1	½	3
4	For 3s state	3	0	0	½	1
5	For 3p state	3	1	0, ± 1	½	3
6	For 3d state	3	2	0, ± 1, ± 2	½	5

c. Now, number of states in shell having Quantum number $1s^2, 2s^2, 2p^6, 3s^2, 3p^6$, i.e., 18.

11. The density of aluminum is 2.70 g/cm³ and its molecular weight is 26.98 g/mole. (a) Calculate the Fermi energy. (b) If the experimental value of E_F is 12 eV, what is the electron effective mass in aluminum? Aluminum is trivalent. **[TU Microsyllabus, P 24.8]**

Solution:

Density of aluminum (ρ) = 2.70 gm cm^{-3}

Molecular weight of Aluminum (m) = 26.98 gm $mole^{-1}$

a. Fermi energy (E_F) = ?

b. Fermi energy (E_F) = 12 eV = $12 \times 1.6 \times 10^{-19}$ J

Effective mass (m_e^*) = ?

We know hat

Number of free electrons per unit volume

$$N = \frac{vrN_A}{m}$$

i.e. $$N = \frac{3 \times 2.70 \times (6.02 \times 10^{23})}{26.98} = 1.807 \times 10^{23} \text{ electrons cm}^{-3}$$

$$= 1.807 \times 10^{29} \text{ electrons m}^{-3}$$

Now, we have relation

$$E_F = \frac{\hbar^2}{2m_e}(3N\pi^2)^{2/3} = \frac{\left(\frac{6.626 \times 10^{33}}{2\pi}\right)^2}{2 \times 9.1 \times 10^{-31}}(3 \times 1.807 \times 10^{29} \times \pi^2)^{2/3}$$

$$= 1.867 \times 10^{-18} \text{ J} = \frac{1.867 \times 10^{-18}}{1.6 \times 10^{-19}} \text{ eV}$$

$\therefore$ $E_F = 11.66$ eV

b. Again, for m_e^*

$$E_F' = \frac{\hbar^2}{2m_e}(3N\pi^2)^{2/3}$$

$$m_e^* = \frac{\hbar^2}{2E_F'}(3N\pi^2)^{2/3}$$

In the diagram, the energy of the electron $E_1 = -13.6$ eV in the ground state, and the values of energy $E_n = \frac{-13.6}{n^2}$ in the excited states for different values of n are plotted on a vertical scale, these energy levels are represented by horizontal lines. As n becomes larger, the energy level becomes progressively closer. For $n = \infty$, $E_\infty = 0$. The energy in eV required to remove the electron from the ground state to infinity i.e. to the zero energy level is called ionization potential or the ionization energy for the hydrogen atom. Its value for the hydrogen atom is 13.6 eV.
The transitions of the electron, which give rise to the Lyman, Balmer, Paschen, Brackett and Pfund series of lines. As n increases the energy difference between successive energy level decreases. Therefore, the separation between the spectral lines of higher frequencies or smaller wavelengths in any series decreases.

5. Give a brief account of Kronig-Penny model.

Solution: [See Model Questions No. 5]

6. Describe the term 'space quantization'

Solution:

An electron revolving about a nucleus is a current loop and has a magnetic field. Let the uniform magnetic field of flux density B and having its direction parallel to z-axis. The magnetic orbital quantum number m_l specifies the direction of L by determining, the component of L in the field direction. The component of L in the z-axis also quantized and it is given by

$$L_z = m_l \hbar$$

Since, for a given l, there are $(2l + 1)$ possible values of

$$m_l (= 0, \pm 1, \pm 2, \ldots \pm l).$$

So, the angular momentum vector L can have $(2l + 1)$ discrete orientations with respect to the magnetic field. This quantization of the orientation of atoms in space is known as 'space quantization'.

The space quantization of the orbital angular momentum of an electron for $l = 2$ is shown in figure.

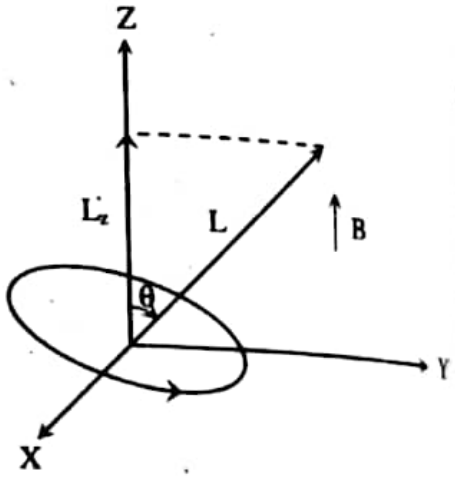
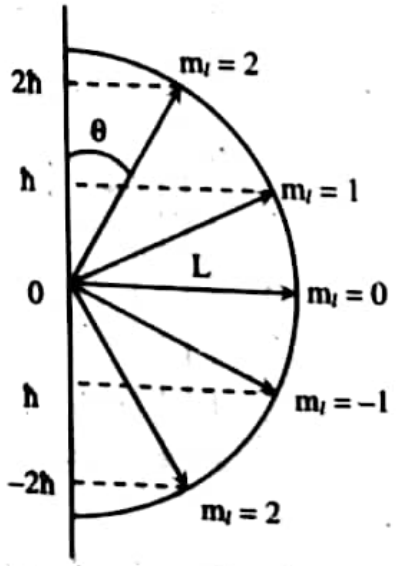


Figure: Electron revolving around nucleus



$$L = \hbar\sqrt{l(l+1)} = \hbar\sqrt{2(2+1)} = \hbar\sqrt{6}$$

Figure: Orientation of angular momentum of electron in hydrogen atom for l = 2

MODEL SET 2

ng Questions

Attempt any TWO questions. [10×2 = 20]

1. Define extrinsic semiconductor. Find an expression for carrier density and fermi level in extrinsic semiconductor.
2. Describe moment of inertia and torque for a rotating rigid body. Find the expression for rotational kinetic energy and discuss the conditions for conservation.
3. Explain degenerate states and Fermi energy on the basis of quantum mechanical free electron theory.

hort Questions

Attempt any EIGHT questions. [5×8=40

4. Derive an expression for the force experienced by a conductor carrying current when placed in a uniform magnetic field.
5. Derive an expression for total energy of an electron revolving around the nucleus in a external magnetic field. Explain.
6. Show that the fermi level in intrinsic semiconductor lies exactly half way between the top of valence band and bottom of the conduction band at absolute zero temperature
7. Suppose the body of an ice skater has a moment of inertia 4 kgm^2 and her arms have a mass 5 kg each with the centre of mass at 0.4 m from her body. She starts to turn at 0.5 rev/sec on the point of her skate with her arms out stretched. She then pulls her arms inward so that their centre of mass is at the axis of her body is zero. What is the speed of rotation? [TU Microsyllabus W 8.4]
   Ans: 0.7 rev/sec.
8. Assume that the electron in a hydrogen atom is essentially in a circular orbit of radius 0.5×10^{-10} m, and rotates about the nucleus at the rate of 10^{14} times per second. What is the magnetic moment of the hydrogen atom due to the orbital motion of the electron? [TU Microsyllabus, W 16.1]
   Ans: 1.26×10^{-25} Am^2
9. An α-particle is emitted from a nucleus with an energy of 5 MeV (5×10^6 eV). Calculate the wavelength of an α-particle with such energy and compare it with the size of the emitting nucleus that has a radius of 8×10^{-15} m. [TU Microsyllabus, P 19.7]
   Ans: 6.409×10^{-15} m
10. For a free Quantum particle show that the wave function $\psi(x, t) = A \cos kxe^{-i\omega t}$ satisfies the time dependent Schrödinger equation. [TU Microsyllabus, P 20.2]
11. Assuming that atoms in a crystalline structure are arranged as close-packed spheres, what is the ratio of the volume of the atoms to the volume available for the simple cubic structure? Assume a one-atom basis. [TU Microsyllabus, P 22.3]
12. Suppose that the effective mass of hole in a material is four times that of electrons. At what temperature would the Fermi level shifted by 10% from the middle of the forbidden energy gap? Let, $E_g = 1$ eV. [TU Microsyllabus, P 25.2]
    Ans: 567.68 K

□□□

The angle θ between the direction of vector L and the z-axis is given by

$$\cos\theta = \frac{L_z}{L} = \frac{m_l h}{h\sqrt{l(l+1)}} = \frac{m_l}{\sqrt{l(l+1)}}$$

A given spring stretches 0.1 m when a force of 20 N pulls on it. A 2-kg block attached to it on a frictionless surface is pulled to the right 0.2 m and released. (a) What is the frequency of oscillation of the block? (b) What is its velocity at the midpoint? (c) What is its acceleration at either end? (d) What are the velocity and acceleration when x = 0.12 m, on the block's first passing this point? [TU Microsyllabus W 10.2]

Solution: [See Model Questions No. 7]

A proton is moving with a velocity $\vec{v} = (3 \times 10^5\,\hat{i} + 7 \times 10^5\,\hat{k})\ ms^{-1}$ in a region where there is a magnetic field $\vec{B} = 0.4\hat{j}T$. What is the force experienced by the proton? [TU Microsyllabus, P 16.12]

Solution:

Velocity of proton $(\vec{v}) = (3 \times 10^5\,\hat{i} + 7 \times 10^5\,\hat{k})\ ms^{-1}$

Magnetic field $(\vec{B}) = 0.4\hat{j}T$

We know, mass of proton $(m_p) = 1.67 \times 10^{-27}$ kg

Charge of proton $(q) = 1.6 \times 10^{-19}$ c

Force experienced by the proton $(\vec{F}) = ?$

We have a relation

$$\vec{F} = q(\vec{v} \times \vec{B})$$
$$= 1.6 \times 10^{-19}\,[(3 \times 10^5\hat{i} + 7 \times 10^5\,\hat{k}) \times 0.4\hat{j}]$$
$$= 1.6 \times 10^{-19}\,[1.2 \times 10^5\,\hat{k} - 2.8 \times 10^5\,\hat{i}]$$
$$= 1.6 \times 10^{-19}\,[1.2\hat{k} - 2.8\hat{i}] \times 10^5$$
$$\vec{F} = [1.92\hat{k} - 4.48\hat{i}] \times 10^{-14}\ N$$

9. **A neutron spectroscopy a beam of mono energetic neutrons is obtained by reflecting reactor neutrons from a beryllium crystal. If separation between the atomic planes of the beryllium crystal is 0.732 Å, what is the angle between the incident neutron beam and the atomic planes that will yield a monochromatic beam of neutrons of wavelength 0.1 Å.** [TU Microsyllabus, P 19.11]

Ans: [See T.U. 2077 Question No. 9]

10. (a) **How many atomic states are there in Hydrogen with n = 3?**
    (b) **How are they distributed among the sub shells? Label each state with appropriate set of Quantum numebers n, l, m_l, m_s.**
    (c) **Show that the number of states in a shell, that is, states having the same n, is given by $2n^2$.**

Solution:

Here,

a. Principle Quantum number (n) = 3.

Number of atomic states $= \frac{n(n+1)}{2} = \frac{3(3+1)}{2} = 6$

MODEL SET 3

Attempt any TWO questions. **[10×2 = 20]**

1. Explain equilibrium current across the pn junction? Use Fermi-Dirae statistics and Maxwell-Boltzmann distribution to show the flow n to p is equal to the flow from p to n. How electron current from p ton (that is, associated with minority carries) is not affected by the height of the potential energy barrier? Explain. [10]
2. Describe moment of inertia and torque for a rotating rigid body. Find the expression for rotational kinetic energy and discuss the conditions for conservation. [10]
3. Explain the theory of black body radiation. Why this theory needs quantum mechanical interpretation? How this interpretation because experimentally successful? Explain. [10]

Group B

Attempt any EIGHT questions. **[5×8=40]**

4. Explain Hall effect and discuss the importance of Hall voltage while manufacturing electronic devices. [5]
5. Discuss effective mass of electrons and holes. [5]
6. Describe electrical conductivity of semiconductors. [5]
7. An oscillating block of mass 250 g takes 0.15 sec to move between the end-points of the motion, which are 40 cm apart (a) What is the frequency of the motion? (b) What is the amplitude of the motion? (c) What is the force constant of the spring. [5]
8. A current of 50A is established in a slab of copper 0.5 cm thick and 2 cm wide. The slab is placed in a magnetic field of 1.5 T. The magnetic field is perpendicular to the plane of the slab and to the current. The free electron in copper is 8.4×10^{28} electrons/m^3. What will the magnitude of Hall voltage across the width of slab? [5]
9. The uncertainty in the position of a particle is equal to the de Broglie wavelength of the particle. Calculate the uncertainty in the velocity of the particle in terms of the velocity of the de Brogie wave associated with the particle. [5]
10. (a) How many atomic states are there in Hydrogen with n = 3?
    (b) How are they distributed among the sub shells? Label each state with appropriate set of Quantum numebers n, l, m_l, m_s.
    (c) Show that the number of states in a shell, that is, states having the same n, is given by $2n^2$.
11. Copper has a face-centered cubic structure with a one-atom basis. The density of copper is 8.96 gm cm^{-3} and its atomic weight is 63.5 g mole^{-1}. What is the length of the unit cube of the structure?

11. The energy gaps of some alkali halides are KCl = 7.6 eV, KBr = 6.3 eV, KI = 5.6 eV. Which of these are transparent to visible light? At what wave length does each become opaque? [TU Microsyllabus, P 24.6]
Ans: 221.7 nm

12. The reverse saturation current of a Silicon diode is $I_0 = 5 \times 10^{-9}$ Å. The voltage across that diode when forward biased is 0.45 V (a) What is the current through the diode at T = 27°C? (b) If the voltage across the diode held constant, and we assume that I_0 doesnot change with temperature. What is the current through the diode at T = 47°C? [TU Microsyllabus, P 26.3]
Ans: (a) 0.178 A (b) 0.06 A

□□□

MODEL SET 5

Long Questions

Attempt any TWO questions. [10×2 = 20]

1. Show that the diode equation for a p-n junction is
$I = I_0\left[e^{\frac{eV_0}{KT}} - 1\right]$, where I_0 is the reverse saturation current and V_0 is the applied voltage.
2. What is SHM? Set up the differential equation of SHM and show that the K.E. and P.E. vary with time but the total energy remains constant.
3. Show that the energy of an electron is quantized in an infinite potential well hence determine the normalized wave function of the electron.

Short Questions

Attempt any EIGHT questions. [5×8=40

4. What is moment of inertia? Derive a relation for kinetic energy of body in angular motion.
5. Explain how Stern-Gerlach experiment describes the space quantization of spin angular momentum.
6. Explain the mechanism of current amplification using common emitter configuration.
7. A grindstone with I = 240 kg -m² rotates with a speed of 2 rev/sec. A knife blade is pressed against it, and the wheel coasts to a stop with constant deceleration in 12 sec. What torque did the knife exert on the wheel? [TU Microsyllabus P 8.4]
Ans: 125.66 Nm
8. An electron is placed midway between two fixed charges, $q_1 = 2.5 \times 10^{-19}$c and $q_2 = 5 \times 10^{-10}$c. If the charges are 1 m apart, what is the velocity of the electron when it reaches a point 10 cm from q_2? [TU Microsyllabus, P 14.21]
Ans: 1.125×106 ms^{-1}
9. After being excited, the electron of a Hydrogen atom eventually falls back to the ground state. This can take place is one jump or in a series of jumps, the electron falling into lower excited states before it end up in the ground state, that is n = 3. Calculate the different photon energies that may be emitted as the atom returns to the ground state. [TU Microsyllabus, W 18.2]
Ans: 10.17 eV
10. What is the probability of finding a particle in a well of width 'a' at a position from the wall if n = 1, if n = 2, if n = 3. Use the normalized wave function, $\psi(x, t) = \sin e^{-iEt/\hbar}$. [TU Microsyllabus, P 20.12]

The output of a digital circuit Y is given by the expression $Y = (B + \bar{C}BA)(A + \bar{C})$, where A, B and C represent inputs. Draw circuit of above equation using OR, AND and NOT gate. Find its truth table.

❑❑❑

MODEL SET 4

Long Questions

Attempt any TWO questions. **[10×2 = 20]**

On the basis of CFE model, derive an expression for thermal conductivity. Also show that the ratio of thermal conductivity to the electrical conductivity is directly proportional to the temperature.

Define electric potential. Find an expression for electric potential due to a point charge. Also find the potential difference between two points.

State the postulates of Bohr's theory. Derive an expression for the frequencies of radiation emitted in hydrogen atom. Hence explain various spectral series of this atom.

Short Questions

Attempt any EIGHT questions. **[5×8=40]**

Calculate the force experienced by a current carrying conductor of length 'l' and cross sectional area 'A' when placed in an uniform magnetic field of strength 'B'.

Write Schrödinger wave equation for hydrogen atom in spherical polar coordinates. Explain degeneracy with the example of hydrogen atom.

Define RS-flip flop. Show that RS flip flop can be constructed by using NAND gate.

A uniform wooden board of mass 20 kg rests on two supports as shown in figure. A 30 kg steel block is placed to the light of support A. How far to the right of a can the steel the block be placed without tipping the board?

[TU Microsyllabus, P 8.7]

Ans: 2m.

8. What force is experienced by a wire of length 0.08 m at an angle of 20° to the magnetic field direction carrying current of 2A in a magnetic field of 1.4T?

[TU Microsyllabus, P 16.1]

Ans: 7.66×10^{-2} N

9. A neutron spectroscopy a beam of mono energetic neutrons is obtained by reflecting reactor neutrons from a beryllium crystal. If separation between the atomic planes of the beryllium crystal is 0.732 Å, what is the angle between the incident neutron beam and the atomic planes that will yield a monochromatic beam of neutrons of wavelength 0.1 Å.

[TU Microsyllabus, P 19.11] **Ans:** 3.92°

10. Show by direct substitution into the time dependent Schrödinger equation for the free particle, that $\psi(x, t) = A\cos(kx - \omega t)$ is not a solution.

[TU Microsyllabus, P 20.1]

11. The density of aluminum is 2.70 g/cm³ and its molecular weight is 26.98 g/mole. (a) Calculate the Fermi energy. (b) If the experimental value of E_F is 12 eV, what is the electron effective mass in aluminum? Aluminum is trivalent [TU Microsyllabus, P 24.8]
Ans: (a) 11.66 eV (b) 0.972 m_e

12. The average saturation current of a Silicon diode doubles when the temperature changes from 27°C to 33°C. What is the position of the fermi level on the pside of the junction. [TU Microsyllabus, P 26.5]
Ans: $E_v - 1.031$ eV

❑❑❑

MODEL SET 7

Long Questions
Attempt any TWO questions. [10×2 = 20]

1. Define n-type and p-type semiconductor? Explain the effect of temperature on the conductivity of n-type and p-type semiconductor.
2. Derive an expression for torque on a current loop in terms of dipole moment and applied magnetic field.
3. State the fundamental postulates of Bohr's theory of hydrogen spectra. Explain the existence of definite energy states on the basis of this theory.

Short Questions
Attempt any EIGHT questions. [5×8=40]

4. State and explain Coulomb's force. Find an expression for electric potential energy for a system of charges?
5. Describe different types of bonding in solids.
6. Why silicon is mostly preferred for semiconducting materials? Explain the zone refining process of purifying the polycrystalline silicon.
7. A large wheel of radius 0.4 m and moment of inertia 1.2 kg m², pivoted at the centre is free to rotate without friction. A rope is wound it and a 2 kg weight is attached to the rope. When the weight has descended 1.5 m from its starting position (a) What is its downward velocity? (b) What is the rotational velocity of wheel? [TU Microsyllabus W 8.2]
Ans: (a) 2.5 m/sec (b) 6.2 rod/sec
8. A proton is accelerated through a potential difference of 200V. If then enters a region where there is a magnetic field B = 0.5T. The magnetic field is perpendicular to the direction of motion of the proton. What is the force experienced by the proton? [TU Microsyllabus, P 16.13] **Ans:** 1.568×10^{-14} N
9. A small particle of mass 10^{-6} g moves along the x-axis, its speed is uncertain by 10^{-6}ms^{-1}.
   a. What is the uncertainty in the x-coordinate of the particle?
   b. Repeat the calculation for an electron assuming that the uncertainty in its velocity is also 10^{-6}ms^{-1}. [TU Microsyllabus, P 19.16]
   Ans: 115.768 m
10. For a free Quantum particle show that the wave function $y(x, t) = A \cos kx\, e^{-iwt}$ satisfies the time dependent Schrodinger equation.

Assuming that atoms in a crystalline structure are arranged as close-packed spheres, what is the ratio of the volume of the atoms to the volume available for the body centered cubic structure? Assume a one-atom basis.

[TU Microsyllabus, P 22.4]

(a) Find the truth table for the circuit of given figure. What logic function does the circuit perform?

(b) What logic function will the circuit perform if the common grounded input to the first two NOR gates are changed to +5V?

[TU Microsyllabus, P 27.10]

❑❑❑

MODEL SET 6

Long Questions **[10×2 = 20]**

Attempt any TWO questions.

What is single crystal growth? Explain the methods of single crystal growth.

Describe moment of inertia and torque of a rigid body. Find the expression for rotational K.E. and discuss the conservation of angular momentum.

Explain, how Franck-Hertz experiment demonstrate the existence of discrete energy level in atoms.

Short Questions **[5×8=40**

Attempt any EIGHT questions.

Explain angular momentum. Show that angular momentum is conserved if there is no applied external torque.

Explain angular momentum. Show that angular momentum is conserved if there is no applied external torque.

Derive Ohm's law on the basis of classical free electron model.

A bicycle wheel of mass 2 kg and radius 0.32 m is spinning freely on its axle at 2 rev sec^{-1}. When you place your hand against the tyre, the wheel decelerates uniformly and comes to a stop in 8 sec. What was the torque of your hand against the wheel?

[TU Microsyllabus P 8.1]

Ans: – 0.64 Nm

A charge $q_1 = 3 \times 10^{-6}$C is located at the origin of x-axis. A second charge $q_2 = -5 \times 10^{-6}$ C is also on the x-axis 4m from the origin in the positive x-direction (a). Calculate the electric field at the mid-point P of the line joining the two charges (b). At what point P' on that line is the resultant field zero?

[TU Microsyllabus, W 14.1] **Ans:** (a) 18×10^3 N/C (b) –1.75

The shortest wavelength of the Paschen series from Hydrogen is 8204Å. From this fact, calculate the Rydberg constant.

[TU Microsyllabus, P 18.3]

Ans: 1.097 × 107 m^{-1}

A beam of hydrogen atoms is used in a Stern Getlach type experiment. The atoms emerge from the oven with a velocity $v = 10^4$ m/sec. They enter a region 20 cm long where there is magnetic field gradient $= 3 \times 10^4$ T/m. The field gradient is perpendicular to the incident velocity of the atoms. The mass of the hydrogen atom is 1.67×10^{-27} kg. What is the separation of the two components of the beam as they emerge from the magnet?

[TU Microsyllabus, W 21.2] **Ans:** 3.33 cm

11. The number of free electrons in copper is 8.4×10^{28} electrons/m^3 (a) Calculate Fermi energy for Cu. (b) At what temperature T_f will the average thermal energy K_BT_f of a gas be equal to that energy?
[TU Microsyllabus, W 23.2]
Ans: (a) 6.95 eV (b) 80,500K

12. The output of a digital circuit Y_1 is given by this expression: $Y = (\bar{A}B + \bar{C}A)(\overline{B + C})$ where A, B and C represent inputs. Draw a circuit of above equation using OR, AND and NOT gate and hence find its truth table.[TU Model 2074]

MODEL SET 8

Long Questions
Attempt any TWO questions. **[10×2 = 20]**

1. Describe Bipolar junction transistor and explain how it is used as voltage amplifier.
2. What is SHM? Set up the differential equation of SHM and show that the K.E. and P.E. vary with time but the total energy remains constant.
3. What is wave function? Explain the physical significances and the characteristics of the wave function.

Short Questions
Attempt any EIGHT questions. **[5×8=40**

4. Derive an expression for the force experienced by a moving charge in the presence of electric and magnetic field.
5. Differentiate insulator, semiconductor and conductor on the basis of band theory.
6. With neat sketch, explain the fabrication of a simple DTL NOR gate.
7. A spring mass system consist of a block of mass 2 kg and a spring with spring constant 200 Nm. The block is released from rest at a position of $x_1 = 0.2$ m (a) What is the velocity at $x_2 = 0.1$ m on other-side of mean position? (b) What is its acceleration at this point? [TU Microsyllabus W 10.3]
Ans: (a) 1.73 m/sec (b) 10 m/sec²
8. A current of 50A is established in a slab of copper 0.5 cm thick and 2 cm wide. The slab is placed in a magnetic field of 1.5 T. The magnetic field is perpendicular to the plane of the slab and to the current. The free electron in copper is 8.4×10^{28} electrons/m^3. What will the magnitude of Hall voltage across the width of slab? [TU Microsyllabus, W 16.2, TU 2074]
Ans: 1.12×10^{-6} V
9. What are (a) the energy, (b) the momentum, and (c) the wavelength of photon that is emitted when a Hydrogen atom undergoes a transition from the state n = 3 to n = 1? (The momentum of the photon is given by).
[TU Microsyllabus, P 18.2, TU Model 2074]
Ans: 6.42×10^{-27} kg ms^{-1}

10. Explain, why the following Eigen function are not acceptable solution of the Schrödinger equation. [TU Microsyllabus, P 20.3]

a. $\chi(x) = 0$ for $x \leq 0$
   $\chi(x) = A\cos kx$ for $x \geq 0$

b. $\chi(x) = A$

c. $\chi = A \ln kx$

11. Assuming that atoms in a crystalline structure are arranged as close-packed spheres, what is the ratio of the volume of the atoms to the volume available for the face centered cubic structure? Assume a one-atom basis. [TU Microsyllabus, P 22.5]

12. The energy gap in germanium is 0.67 eV. The electron and the hole effective masses are 0.12 m and 0.23 m respectively, where m in the free electron mass. Calculate (a) the fermi energy (b) the electron density and (c) the hole density at T = 300 K. [TU Microsyllabus, P 25.3]

Ans: (a) 0.348 eV (b) $3.96 \times 10^{18}\ m^{-3}$ (c) $1.05 \times 10^{19}\ m^{-3}$

❑❑❑